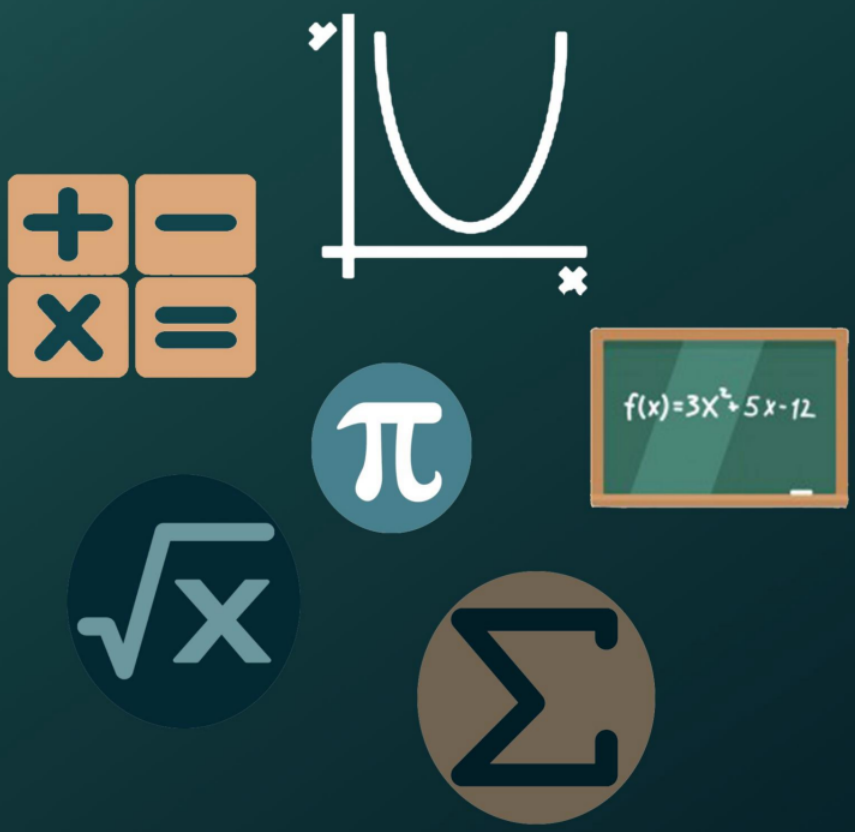
MATHEMATICS

Solution of Triangle



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Properties & Solution of Triangle	1-3		1-3	
2	Sine Rule	4-8	1, 10	4-7	
3	Cosine Rule	9-13	11		1
4	Projection Formula	14-16	2, 12	8-14	
5	Tangent Rule	17-18	13	15	2, 19
6	Half Angle Formulae	19-21		16-18	3
7	m-n Theorem	22			17
8	Circumcircle	23-24			
9	Incircle		7, 9, 14	22	20
10	Excircles	25-28	4-6, 15	23-24	4, 21
11	Orthocentre	29			
12	Lengths of Angle Bisector & Median	30-31		19, 25	5
13	Regular polygon			26-27	11-13
14	After Completing Chapter		3, 8, 16-19	20-21, 28	6-10, 14-16, 18, 22-25

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Properties & Solution of Triangle

Q1. If in ΔABC, $\sin B \cos C = \frac{\sqrt{3}-1}{\sqrt{3}}$ and $\cos B \sin C = \frac{1}{\sqrt{3}}$, then the triangle is -

(A) Isosceles
(B) Equilateral
(C) Right angled
(D) Right angled isosceles

Q2. In a ΔABC if $\frac{\tan A}{1} = \frac{\tan B}{2} = \frac{\tan C}{3}$, then $\angle A$ is equal to -

(A) $\frac{\pi}{12}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{3}$

Q3. P is point inside an equilateral triangle. If distacne of P from sides 4,5 and 6 then length of sides of the triangle is -

(A) $5\sqrt{3}$ (B) $10\sqrt{3}$
(C) $15\sqrt{3}$ (D) $20\sqrt{3}$

Sine Rule

Q4. If the angles of a triangle are in the ratio 1 : 2 : 3 then their corresponding sides are in the ratio.

(A) $1:\sqrt{3}:3$ (B) $1:\sqrt{2}:3$
(C) $1:\sqrt{3}:2$ (D) None

Q5. In ΔABC, $\angle A = 45°$, $\angle C = 60°$ then $a + c\sqrt{2}$ is equal to

(A) b (B) 2b
(C) $\sqrt{2}\,b$ (D) $\sqrt{3}\,b$

Q6. In a ΔABC with usual notations, if $\frac{a}{\cos A} = \frac{b}{\cos B} = \frac{c}{\cos C}$, then the triangle is-

(A) right angled (B) isosceles
(C) obtuse angled (D) equilateral

Q7. In ΔABC, $c \cos(A - \alpha) + a \cos(C + \alpha)$

(A) $a \cos \alpha$ (B) $b \cos \alpha$
(C) $c \cos \alpha$ (D) $b \sin \alpha$

Q8. If in a triangle ABC, $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$, then the triangle is -

(A) Right angled or isosceles
(B) Right angled and isosceles
(C) Equilateral
(D) None of these

Cosine Rule

Q9. If the sides of a triangle are 3, 5 and 7 then triangle is :

(A) Right angled (B) Obtuse angled
(C) Acute angled (D) None

Q10. If the lengths of the sides of a triangle be $7, 4\sqrt{3}$ and $\sqrt{13}$ cm, then the smallest angle is :

(A) 15° (B) 30° (C) 60° (D) 45°

Q11. In ΔABC if $(a + b + c)(a - b + c) = 3ac$, then find $\angle B$

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q12. If a = 9, b = 8 and c = x satisfies 3cos C = 2, then -

(A) x = 5 (B) x = 6

(C) x = 4 (D) x = 7

Q13. The expression

$$\frac{(a+b+c)(b+c-a)(c+a-b)(a+b-c)}{4b^2c^2}$$

is equal to -

(A) $\cos^2 A$ (B) $\sin^2 A$

(C) cosA cosB cosC (D) None of these

Projection Formula

Q14. Value of $\frac{\cos C+\cos A}{c+a}+\frac{\cos B}{b}$ is -

(A) $\frac{1}{b}$ (B) $\frac{2}{b}$ (C) $\frac{3}{b}$ (D) $\frac{4}{b}$

Q15. The value of

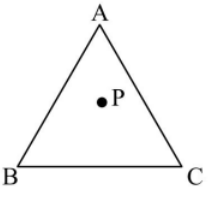
$$\left(\frac{b}{c}+\frac{c}{b}\right)\cos A+\left(\frac{a}{b}+\frac{b}{a}\right)\cos C+\left(\frac{a}{c}+\frac{c}{a}\right)\cos B$$

is equal to -

(A) 6 (B) 3

(C) 4 (D) 5

Q16. In the adjacent figure 'P' is any interior point of the equilateral triangle ABC of side length 2 unit.



If x_a, x_b and x_c represent the distance of P from the sides BC, CA and AB respectively then $x_a + x_b + x_c$ is equal to -

(A) 6 (B) $\sqrt{3}$

(C) $\frac{\sqrt{3}}{2}$ (D) $2\sqrt{3}$

Tangent Rule

Q17. In a triangle ABC, $\angle A = 60°$ and $b : c = \sqrt{3} + 1 : 2$ then value of $(\angle B - \angle C)$ is -

(A) 15° (B) 30°

(C) 45° (D) 90°

Q18. In a ΔABC, $\tan\left(\frac{A-B}{2}\right)\cot\left(\frac{A+B}{2}\right)$ is equal to -

(A) $\frac{b-a}{b+a}$ (B) $\frac{a-b}{a+b}$

(C) $\frac{c+a}{c-a}$ (D) None of these

Half Angle Formulae

Q19. In a ΔABC, with usual notations if $c^2 = a^2 + b^2$, then $4s(s-a)(s-b)(s-c) =$

(A) s^4 (B) c^2b^2

(C) s^2c^2 (D) a^2b^2

Q20. If in a triangle ABC, 3a = b + c then $\cot\frac{B}{2}\cdot\cot\frac{C}{2}$ is equal to -

(A) 1 (B) 2

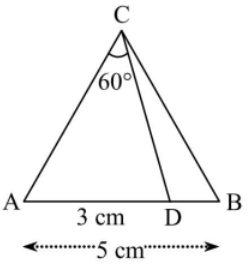
(C) 3 (D) $\sqrt{3}$

Q21. If area of triangle is 75 sq. meters and two of its sides are 20 m and 15 m, then included angle is-

(A) $\frac{\pi}{6}$ OR $\frac{5\pi}{6}$ (B) $\frac{\pi}{3}$ OR $\frac{2\pi}{3}$

(C) $\frac{\pi}{4}$ OR $\frac{\pi}{2}$ (D) None of these

Q22. In the figure, ABC is triangle in which $C = 90^\circ$ and $AB = 5$ cm. D is a point on AB such that $AD = 3$ cm and $\angle ACD = 60^\circ$. Then the length of AC is -



(A) $5\sqrt{\frac{3}{7}}$ cm (B) $\sqrt{\frac{3}{7}}$ cm

(C) $\frac{3}{\sqrt{7}}$ cm (D) none of these

Circumcircle

Q.23 If Δ stands for the area of a triangle ABC then $a^2\sin2B + b^2\sin2A$ is equal to -

(A) Δ (B) 2Δ

(C) 3Δ (D) 4Δ

Q24. If in a ΔABC, $b = 2$, $C = \sqrt{3}$, $\angle A = \frac{\pi}{6}$, then R = ??

(A) 1/2 (B) 2

(C) 1 (D) None

Excircles

Q25. In any ΔABC $\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}$ is equal to $\{r_1, r_2, r_3$ are exradii$\}$

(A) $\frac{\Delta}{s}$ (B) $\frac{s}{\Delta}$

(C) $\frac{2s}{\Delta}$ (D) None of these

Q26. If $\frac{r}{r_1}=\frac{r_2}{r_3}$, then -

(A) $\angle A = 90^\circ$ (B) $\angle B = 90^\circ$

(C) $\angle C = 90^\circ$ (D) None of these

Q27. If in triangle ABC (with usal notations), $r_1 > r_2 > r_3$, then -

(A) $a < b < c$ (B) $a < c < b$

(C) $a > b > c$ (D) $a > c > b$

Q28. In equilateral triangle, ratio of inradius, cirum radii and exradii is -

(A) 2 : 3 : 5 (B) 1 : 2 : 3

(C) 3 : 7 : 9 (D) 3 : 7 : 8

Orthocentre

Q29. For an acute angle ΔABC, $a = 10$, $\angle A = \frac{\pi}{6}$ & H is orthocenter then circumradius of triangle formed by H, B & C is -

(A) 5 (B) 10

(C) 20 (D) 40

Lengths of Angle Bisector & Median

Q30. The length of medians of triangle are 3, 4, 5 units, then area of triangle in square units, is -

(A) 4 (B) 6

(C) 7 (D) 8

Q31. In ΔABC, if $AC = 4$, $AB = 5$ and $\angle A = \frac{2\pi}{3}$, then the length of internal angle bisector through A is -

(A) $\frac{20}{9}$ (B) $\frac{10}{9}$

(C) $\frac{9}{10}$ (D) $\frac{9}{20}$

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If the sines of the angles A and B of a triangle ABC satisfy the equation $c^2x^2 - c(a+b)x + ab = 0$, then the triangle-

(A) is acute angled

(B) is right-angled

(C) is obtuse angled

(D) satisfies $\sin A + \cos A = (a+b)/c$

Q2. In ΔABC, $b = 3$, $c = 4$ and $\angle A = \frac{\pi}{3}$, then select the correct options :

(A) Area of ΔABC is $3\sqrt{3}$ sq. units

(B) Possible value of a is $\sqrt{13}$

(C) Area of ΔABC is $6\sqrt{3}$ sq. units

(D) Possible value of a is $\sqrt{15}$

Q3. In a triangle ABC if circumcentre, centroid and orthocentre are coinciding (here a, b, c are the sides of triangle opposite to vertices A, B & C respectively) and $a = 2$, then -

(A) value of $\sin^2A + \cos^2B + \tan^2C = 4$

(B) area of $\Delta ABC = \sqrt{3}$

(C) inradius of $\Delta ABC = \frac{1}{\sqrt{3}}$

(D) distance of orthocentre of ΔABC from the vetex A is $\frac{2}{\sqrt{3}}$.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 4 to 6)

Given that $\Delta = 6$, $r_1 = 2$, $r_2 = 3$, $r_3 = 6$

Q4. Circumradius R is equal to-

(A) 2.5 (B) 3.5

(C) 1.5 (D) none of these

Q5. Inradius is equal to-

(A) 2 (B) 1

(C) 1.5 (D) 2.5

Q6. Difference between the greatest and the least angle is-

(A) $\cos^{-1}\frac{4}{5}$ (B) $\tan^{-1}\frac{3}{4}$

(C) $\cos^{-1}\frac{3}{4}$ (D) $\cos^{-1}\frac{3}{5}$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q7. **Match the following :**

	Column- I	**Column- II**
	In a triangle ABC	
(A)	If a, b, c are 13, 14, 15 respectively than r_1 is equal to	(P) 8
(B)	The inradius of triangle whose sides are 3, 5, 6 is 'r' then $7r^2$ is equal to	(Q) 10.5
(C)	If the radius of circumcircle of an isosceles triangle ABC is equal to AB (where AB = AC), then angle $A = \frac{p\pi}{3}$, then 4p is equal to	(R) 10
(D)	In an equilateral triangle , $\frac{5R}{r}$ is equal to	(S) 7

Q8. Match the following :

	Column-I	**Column-II**
(A)	In a triangle ABC, if a is the arithmetic mean and b, c ($b \neq c$) are the two geometric means between any two positive real numbers then $\frac{\sin^3 B + \sin^3 C}{\sin A \sin B \sin C}$ is equal to	(P) 9
(B)	In a triangle ABC; $\frac{(a^2 - b^2 - c^2)\tan A + (a^2 + c^2 - b^2)\tan B}{2}$ is equal to :-	(Q) 2
(C)	In a triangle ABC, if $B = 30^\circ$ and $c = \sqrt{3}\, b$, then $\frac{A}{45^\circ}$ can be equal to	(R) 1
(D)	In a ΔABC, if $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then Σ 4 sin A sin B =	(S) 0

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q9. If radius of the incircle of a triangle with its sides 5p, 6p and 5p is 6, then p is equal to ?

SUBJECTIVE TYPE

Q10. Prove that in a ΔABC

(i) $$\frac{\cos^2 B - \cos^2 C}{b + c} + \frac{\cos^2 C - \cos^2 A}{c + a} + \frac{\cos^2 A - \cos^2 B}{a + b} = 0$$

(ii) $a^2 (\cos^2 B - \cos^2 C) + b^2 (\cos^2 C - \cos^2 A) + c^2 (\cos^2 A - \cos^2 B) = 0$

(iii) $\frac{a^2\sin(B-C)}{\sin B+\sin C}+\frac{b^2\sin(C-A)}{\sin C+\sin A}+\frac{c^2\sin(A-B)}{\sin A+\sin B}=0$

(iv) $(b^2-c^2)\cot A+(c^2-a^2)\cot B+(a^2-b^2)\cot C=0$

(v) $\left(\frac{b^2-c^2}{a^2}\right)\sin 2A+\left(\frac{c^2-a^2}{b^2}\right)\sin 2B+\left(\frac{a^2-b^2}{c^2}\right)\sin 2C=0$

Q11. (i) The equation $ax^2+bx+c=0$, where a, b, c are the sides of a triangle ABC and the equation $x^2+\sqrt{2}x+1=0$ have a common root, then find the value of $\angle C$.

(ii) The equation $ax^2+bx+c=0$, where a, b, $c\in N$, are the sides of a triangle ABC and the equation $x^2+2x+3=0$ have a common root, then find the minimum value of $(a+b+c)$.

(iii) If a, b, c are the sides of triangle ABC and a^2, b^2, c^2 are roots of $x^3-Px^2+Qx-R=0$, where P, Q, R are constants, then show that

$$\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}=\frac{P}{2\sqrt{R}}.$$

Q12. Prove that in ΔABC

(i) $\frac{\cos B}{\cos C}=\frac{c-b\cos A}{b-c\cos A}$

(ii) $\frac{\cos A}{c\cos B+b\cos C}+\frac{\cos B}{a\cos C+c\cos A}+\frac{\cos C}{a\cos B+b\cos A}=\left(\frac{a^2+b^2+c^2}{2abc}\right)$

Q13. If in a ΔABC, we define $x=\tan\left(\frac{B-C}{2}\right)\tan\left(\frac{A}{2}\right)$, $y=\tan\left(\frac{C-A}{2}\right)\tan\left(\frac{B}{2}\right)$ and $z=\tan\left(\frac{A-B}{2}\right)\tan\left(\frac{C}{2}\right)$.

Then show that $x+y+z=-xyz$

Q14. If incenter of ΔABC (with usual notations) is I and $IA\times IB\times IC=KRr^2$, then the value of 'k' is -

Q15. Prove that in a triangle ABC.

(i) $R\,r\,(\sin A+\sin B+\sin C)=\Delta$

(ii) $(r+r_1)\tan\left(\frac{B-C}{2}\right)+(r+r_2)\tan\left(\frac{C-A}{2}\right)+(r+r_3)\tan\left(\frac{A-B}{2}\right)=0$

(iii) $\frac{bc-r_2r_3}{r_1}=\frac{ca-r_3r_1}{r_2}=\frac{ab-r_1r_2}{r_3}=r$

(iv) $\frac{\sum\cot\left(\frac{A}{2}\right)}{\sum\cot A}=\frac{(a+b+c)^2}{(a^2+b^2+c^2)}$

Q16. (i) The ratio of the circum-radius and the in-radius of ΔABC, whose sides are in the ratio 4 : 5 : 6, is expressed as $\left(\frac{m}{n}\right)$; m, $n\in N$, H.C.F. (m, n) = 1. Then find $(m-n)$.

(ii) For equilateral triangle, prove that $r : R : r_1 = 1 : 2 : 3$.

(iii) If $r_1 : r_2 : r_3 = 1 : 2 : 3$ then prove that $a : b : c = 5 : 8 : 9$.

Q17. Let ABC be a triangle.

(i) Suppose that area of $\Delta ABC = 10\sqrt{3}$, $|AB| = 8$ and $|AC| = 5$. Find all possible values $\angle A$.

(ii) Suppose that $|AB| = 5\sqrt{2}$, $|BC| = 5\sqrt{3}$ and $\angle C = 45^\circ$. Find all possible values of $\angle A$.

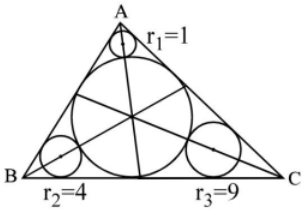
(iii) Suppose that $|AB| = 5\sqrt{2}$, $|BC| = 5$ and $\angle C = 45^\circ$. Find all possible values of $\angle A$.

(iv) Suppose that $|AB| = 5\sqrt{2}$, $|BC| = 10$ and $\angle C = 45^\circ$. Find the possible values of $\angle A$.

(v) Suppose that $|AB| = 5\sqrt{2}$, $|BC| = 15$ and $\angle C = 45^\circ$. Find all possible values of $\angle A$.

Q18. ABCD is a rhombus. The circumradii of Δ ABD and Δ ACD are 12.5 and 25 respectively. Find the area of rhombus.

Q19. ABC is a triangle. Circles with radii as shown are drawn inside the triangle each touching two sides and the incircle. Find the radius of the incircle of the ΔABC.



Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q.1 The angles of a triangle are in the ratio 1 : 2 : 7. If the ratio of the greatest side to least side is $\frac{\cos(36^\circ)}{\sin(k^\circ)}$, then the value of $\cot(5k^\circ)$ is equal to -

(A) 0 (B) 1

(C) $\sqrt{2}+1$ (D) $2+\sqrt{3}$

Q2. If in a triangle ABC, $\sin A = \sin^2 B$ and $2\cos^2 A = 3\cos^2 B$, then the Δ ABC is -

(A) right angled (B) obtuse angled

(C) isosceles (D) equilateral

Q3. In a ΔABC,

$\cos B.\cos C + \sin B.\sin C \sin^2 A = 1$.

Then the triangle is-

(A) right-angle isosceles

(B) isosceles whose equal angles are greater than $\pi/4$

(C) equilateral

(D) none of these

Q4. In a triangle ABC, $\frac{\cos\frac{B-C}{2}}{\sin\frac{A}{2}}$ is equal to :-

(A) $\frac{b-c}{a}$ (B) $\frac{a}{b-c}$

(C) $\frac{b+c}{a}$ (D) $\frac{a}{b+c}$

Q5. In a triangle ABC, right angled at C, the value of cotA + cotB is :-

(A) $\frac{c^2}{ab}$ (B) a + b

(C) $\frac{a^2}{bc}$ (D) $\frac{b^2}{ac}$

Q6. Points D, E are taken on the side BC of a triangle ABC such that BD = DE = EC.

If $\angle BAD = x$, $\angle DAE = y$, $\angle EAC = z$,

then the value of $\frac{\sin(x+y)\sin(y+z)}{\sin x \sin z} =$

(A) 1

(B) 2

(C) 4

(D) none

Q7. If the base angles of a triangle are 22.5° and 112.5°, the height of the triangle is equal to-

(A) half the base

(B) the base

(C) twice the base

(D) four times the base

Q8. In ΔABC, if $a^2\cos 2A = 2b^2 + 2c^2 - a^2$, then A belongs to -

(A) $\left(0, \frac{\pi}{6}\right)$ (B) $\left(\frac{\pi}{6}, \frac{\pi}{4}\right)$

(C) $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (D) $\left(\frac{\pi}{2}, \pi\right)$

Q9. In a ΔABC if $2c^2 + b^2 - 2bc = 6ac - 9a^2$, (with usual notation), then value of cosB is-

(A) $\frac{1}{2}$ (B) $\frac{1}{3}$

(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

Q10. If two adjacent sides of a cyclic quadrilateral are 2 and 5 and the angle between them is 60°. If the third side is 3, then the fourth side of rhombus is :-

(A) 2 (B) 3

(C) 4 (D) 5

Q11. In a Δ ABC, if sides a, b, c are the roots of the equation $x^3 - 11x^2 + 38x - 40 = 0$; then

$$\frac{\cos A}{a} + \frac{\cos B}{b} + \frac{\cos C}{c} =$$

(A) 1 (B) $\frac{3}{4}$

(C) $\frac{9}{16}$ (D) $\frac{16}{9}$

Q12. In ΔABC, $a^2 + b^2 + c^2 = ac + ab\sqrt{3}$, then the triangle is -

(A) equilateral (B) isosceles

(C) right angled (D) none of these

Q13. If two sides of an acute angled triangle are 6 & 7, then number of possible integral values of length of third side is -

(A) 3 (B) 4 (C) 6 (D) 9

Q14. If $\sin\theta$ and $-\cos\theta$ are the roots of the equation $ax^2 - bx - c = 0$, where a,b,c are the sides of a ΔABC, then cosB is equal to -

(A) $1 - \frac{c}{2a}$ (B) $1 - \frac{c}{a}$

(C) $1 + \frac{c}{2a}$ (D) $1 + \frac{c}{3a}$

Q15. If one side of triangle is double the other, and the angles opposite to these sides differ by 60°, then the triangle is-

(A) equilateral (B) obtuse angled

(C) right angled (D) acute angled

Q16. In ΔABC, $\Delta = a^2 - (b-c)^2$, then value of tanA :-

(A) $\frac{4}{15}$ (B) $\frac{6}{15}$

(C) $\frac{8}{15}$ (D) None of these

Q17. In ΔABC, $8\Delta = (b + c)(bc + 1)$, then circumradius of ΔABC is (Δ denotes area of triangle and b, c are length of sides AC and AB respectively) -

(A) $\sqrt{\Delta}$ (B) $\frac{1}{\sqrt{2\Delta}}$

(C) $\sqrt{2\Delta}$ (D) $\frac{1}{\sqrt{\Delta}}$

Q18. In ΔABC with usual notations, if a,b,c are in AP, then the value of $\sqrt{3}\tan\frac{A}{2}\tan\frac{C}{2}$ equals-

(A) 1 (B) $\sqrt{3}$

(C) $\frac{1}{\sqrt{3}}$ (D) $2\sqrt{3}$

Q19. In a ΔABC if median AD is perpendicular to AB, then $\tan A + 2\tan B$ is equal to -

(A) 1 (B) 3

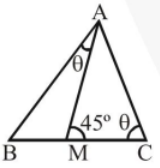
(C) 0 (D)1/2

Q20. In ΔABC, $\angle ABC = 45°$, point D is on BC such that $2BD = CD$ and $\angle DAB = 15°$. The value of $\angle ACB$ is equal to -

(A) $\frac{\pi}{3}$ (B) $\frac{5\pi}{12}$

(C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Q21. If median AM from vertex A on opposite side makes angle θ with side AB (as shown in figure), then $\tan^2\theta$ is -



(A) 1 (B) $\frac{1}{2}$

(C) 3 (D) $\frac{1}{3}$

Q22. In a right angled isosceles triangle the ratio of the circumradius and inradius is-

(A) $2\left(\sqrt{2}+1\right):1$ (B) $\left(\sqrt{2}+1\right):1$

(C) 2 : 1 (D) $\sqrt{2}:1$

Q23. In ΔABC, least value of $\frac{r_1+r_2+r_3}{r}$ is :-

(A) 9 (B) 27

(C) 81 (D) 3

Q24. Let r_1, r_2, r_3 are ex-radii of triangle ABC. If $r_1 = 10$, $r_2 = 15$, $r_3 = 30$ then area of triangle will be-

(A) 100 sq. unit (B) 150 sq. unit

(C) 200 sq. unit (D) 250 sq. unit

Q25. If f,g,h are length of internal bisector of ΔABC, from vertices A, B, C respectively then $\frac{1}{f}\cos\frac{A}{2}+\frac{1}{g}\cos\frac{B}{2}+\frac{1}{h}\cos\frac{C}{2}$ is equal to

(A) $\frac{1}{a}+\frac{1}{b}+\frac{1}{c}$ (B) $\frac{1}{a}+\frac{1}{b}-\frac{1}{c}$

(C) $\frac{1}{a}-\frac{1}{b}+\frac{1}{c}$ (D) $\frac{1}{b}+\frac{1}{c}-\frac{1}{a}$

Q26. A piece of paper is in the shape of a square of side 1m long. It is cut at the four corners to make a regular polygon of eight sides (octagon). The area of the polygon is -

(A) $2\left(\sqrt{2}-1\right)m^2$ (B) $\left(\sqrt{2}-1\right)m^2$

(C) $\frac{1}{\sqrt{2}}m^2$ (D) none of these

Q27. The ratio of area of a regular polygon of n sides inscribed in a circle to that of the polygon of same number of side circum scribe the same circle is 3 : 4, then value of 'n' is :-

(A) 1 (B) 8

(C) 12 (D) 6

Q28. Given an isosceles triangle, whose one angle is 120° and radius of its incircle is $\sqrt{3}$. Then the area of triangle in sq. units is :-

(A) $7+12\sqrt{3}$ (B) $12-7\sqrt{3}$

(C) $12+7\sqrt{3}$ (D) 4π

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. In a triangle ABC is
$2a^2b^2 + 2b^2c^2 = a^4 + b^4 + c^4$,
then angle B is equal to-
(A) 45° (B) 135°
(C) 120° (D) 60°

Q2. In a ΔABC, A = 60° and b : c = 2 : 3.
If $\tan\alpha = \frac{\sqrt{3}}{5}, 0 < \alpha < \frac{\pi}{2}$, then-
(A) $B = 60° + \alpha$ (B) $C = 60° + \alpha$
(C) $B = 60° - \alpha$ (D) $C = 60° - \alpha$

Q3. In triangle ABC if sides a,b,c are in AP and sum of reciprocals of lengths of altitudes is 1, then length of altitude through vertex B can not be -
(A) 3 (B) 2
(C) 4 (D) 5

Q4. Let ABC be a triangle with altitudes h_1, h_2, h_3 and in-radius r.
Then, $\frac{h_1+r}{h_1-r} + \frac{h_2+r}{h_2-r} + \frac{h_3+r}{h_3-r}$ is always greater than?
(A) 7 (B) 6
(C) 5 (D) 4

Q5. In triangle ABC, the length of the bisector of angle A is
(A) $\frac{2bc\sin(A/2)}{b+c}$
(B) $\frac{2bc\cos(A/2)}{b+c}$
(C) $\frac{abc}{2R(b+c)}\operatorname{cosec}\frac{A}{2}$
(D) $\frac{4\Delta}{b+c}\operatorname{cosec}\frac{A}{2}$

Q6. In a ΔABC, tan A and tan B satisfy the inequation $\sqrt{3}x^2 - 4x + \sqrt{3} < 0$. Then-
(A) $a^2 + b^2 + ab > c^2$ (B) $a^2 + b^2 - ab < c^2$
(C) $a^2 + b^2 > c^2$ (D) none of these

Q7. Which of the following pieces of data does NOT uniquely determine an acute-angled triangle ABC(R being the radius of the circumcircle)?
(A) a, sinA, sinB (B) a, b, c
(C) b, sinB, R (D) a, sinA, R

Q8. The ratio of the sides of a triangle ABC is $1 : \sqrt{3} : 2$. The ratio A : B : C is -
(A) 3 : 5 : 2 (B) $1 : \sqrt{3} : 2$
(C) 3 : 2 : 1 (D) 1 : 2 : 3

Q9. In ΔABC, a, b, c are the lengths of its sides and A, B, C are the angles of triangle ABC. The correct relation is/are :-
(A) $(b-c)\sin\left(\frac{B-C}{2}\right) = a\cos\left(\frac{A}{2}\right)$
(B) $(b-c)\cos\left(\frac{A}{2}\right) = a\sin\left(\frac{B-C}{2}\right)$
(C) $(b+c)\sin\left(\frac{B+C}{2}\right) = a\cos\left(\frac{A}{2}\right)$
(D) $(b-c)\cos\left(\frac{A}{2}\right) = 2a\sin\left(\frac{B+C}{2}\right)$

Q10. Internal bisector of $\angle A$ of a triangle ABC meets side BC at D. A line drawn through D perpendicular to AD intersects the side AC at E and the side AB at F. If a, b, c represent sides of ΔABC then

(A) AE is HM of b and c

(B) $AD = \frac{2bc}{b+c}\cos\frac{A}{2}$

(C) $EF = \frac{4bc}{b+c}\sin\frac{A}{2}$

(D) the triangle AEF is isosceles

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 11 to 13)

Let A_n be the area that is outside a n-sided regular polygon and inside it's circumscribing circle. Also B_n is the area inside the polygon and outside the circle inscribed in the polygon. Let R be the radius of the circle circumscribing n-sided polygon.

Q11. If n = 6 then A_n is equal to-

(A) $R^2\left(\frac{\pi-\sqrt{3}}{2}\right)$ (B) $R^2\left(\frac{2\pi-6\sqrt{3}}{2}\right)$

(C) $R^2\left(\pi-\sqrt{3}\right)$ (D) $R^2\left(\frac{2\pi-3\sqrt{3}}{2}\right)$

Q12. If n = 4 then B_n is equal to -

(A) $R^2\frac{(4-\pi)}{2}$ (B) $R^2\frac{(4-\pi\sqrt{2})}{2}$

(C) $R^2\frac{(4\sqrt{2}-\pi)}{2}$ (D) none of these

Q13. $\frac{A_n}{B_n}$ is equal to $\left(\theta=\frac{\pi}{n}\right)$-

(A) $\frac{2\theta-\sin 2\theta}{\sin 2\theta-\theta\cos^2\theta}$

(B) $\frac{2\theta-\sin\theta}{\sin 2\theta-\theta\cos^2\theta}$

(C) $\frac{\theta-\cos\theta\sin\theta}{\cos\theta\,(\sin\theta-\theta\cos\theta)}$

(D) none of these

Paragraph # 2 (Ques. 14 to 16)

In a triantgle if the sum of two sides is x and this product is $y\left(x \geq 2\sqrt{y}\right)$ such that (x + z) (x – z) = y where z is the third side of the triangle.

Q14. Greatest angle of the triangle is -

(A) 105° (B) 120°

(C) 135° (D) 150°

Q15. Inradius of the triangle is

(A) $\frac{\sqrt{3}\,y}{(z+x)}$ (B) $\frac{z}{2(x+y)}$

(C) $\frac{y\sqrt{3}}{2(z+x)}$ (D) $\frac{z\sqrt{3}}{2(x+y)}$

Q16. Area of the triangle is

(A) $\frac{y\sqrt{3}}{4}$ (B) $\frac{x\sqrt{3}}{4}$

(C) $\frac{z\sqrt{3}}{4}$ (D) None of these

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q17. In a triangle ABC, D and E are points in BC such that $\angle BAD = \angle DAE = \angle EAC = A/3$ and $AD = 1$.

Column - I	**Column - II**
(A) BD	(P) $\sin(B + A/3)/\sin(C + A/3)$
(B) DE	(Q) $\sin(A/3)\sin(B + A/3)/\sin C \sin(C + A/3)$
(C) EC	(R) $\sin(A/3) / \sin(C + A/3)$
(D) AE	(S) $\sin(A/3) / \sin B$

Q18. If p_1, p_2, p_3 are altitudes of a triangle ABC from the vertices A, B, C respectively and Δ is the area of the triangle and s is semi perimeter of the triangle, then match the columns

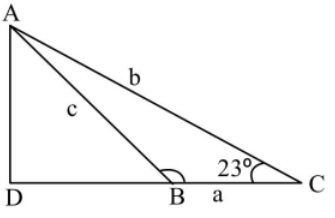
Column-I	**Column-I**
(A) If $\frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{2}$ then the least value of $p_1p_2p_3$ is	(P) $\frac{1}{R}$
(B) The value of $\frac{\cos A}{p_1}+\frac{\cos B}{p_2}+\frac{\cos C}{p_3}$ is	(Q) 216
(C) The minimum value of $\frac{b^2p_1}{c}+\frac{c^2p_2}{a}+\frac{a^2p_3}{b}$ is	(R) 6Δ
(D) The value of $p_1^{-2}+p_2^{-2}+p_3^{-2}$ is	(S) $\frac{\Sigma a^2}{4\Delta^2}$

SUBJECTIVE TYPE

Q19. In a triangle ABC, AD is the altitude from A (fig.).

Given $b > c$, $\angle C = 23°$ and $AD = \frac{abc}{b^2 - c^2}$

then $\cos(B - 23°)$ is equal to -



Q20. The sides of a right angle triangle ABC are all integers & the smallest side is 11. Then the value of inradius of triangle ABC is -

Q21. In a triangle ABC, let $2r_1 = r_2 = r_3$ and $b = 3$. If area of triangle is λ, then value of λ^2 is - (All symbols used regarding triangle ABC has usual meaning)

Q22. If Δ is the area of a triangle with side lengths a, b, c, then show that:

$$\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$$

Also show that equality occurs in the above inequality if and only if $a = b = c$.

Q23. If I_n is the area of n sided regular polygon inscribed in a circle of unit radius and O_n be the area of the polygon circumscribing the given circle, prove that :

$$I_n = \frac{O_n}{2}\left(1+\sqrt{1-\left(\frac{2I_n}{n}\right)^2}\right)$$

Q24. Circles with radii 3, 4 and 5 touch each other externally if P is the point of intersection of tangents to these circles at their points of contact. Find the distance of P from the points of contact.

Q25. Let ABC and ABC′ be two non-congruent triangles with sides $AB = 4, AC = AC' = 2\sqrt{2}$ and angle B = 30°. The absolute value of the difference between the areas of these triangles is -

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9	10
C	C	B	C	B	D	B	A	B	B
11	12	13	14	15	16	17	18	19	20
B	D	B	A	B	B	B	B	D	B
21	22	23	24	25	26	27	28	29	30
A	A	D	C	A	C	C	B	B	D
31									
A									

Ex - 1A

1. B,D **2.** A,B **3.** A,B,C,D **4.** A **5.** B

6. D

7. A → Q ; B → P ; C → P ; D → R

8. A → Q, B → S, C → Q, D → P

9. 4 **11.** (i) 45°, (ii) 6 **14.** 4 **16.** (i) 9

17. (i) ∠A = 60° or 120°, (ii) ∠A = 60° or 120°; (iii) ∠A = 30°,

(iv) ∠A = 90°, (v) Triangle is not possible

18. 400 unit^2 **19.** 11

Ex - 2									
1	2	3	4	5	6	7	8	9	10
A	B	A	C	A	C	A	D	D	A
11	12	13	14	15	16	17	18	19	20
C	C	C	C	C	C	A	C	C	B
21	22	23	24	25	26	27	28		
D	B	A	B	A	A	D	C		

Ex - 2A

1. (A,B) **2.** [B, C] **3.** (B,C,D) **4.** (C, D) **5.** B,C

6. [A, B] **7.** [C, D] **8.** D **9.** B **10.** A,B,C,D

11. D **12.** A **13.** C **14.** (B) **15.** (C)

16. A

17. (A - S) (B - R) (C - Q) (D - P)

18. (A) $\rightarrow$ (Q), (B) $\rightarrow$ (P), (C) $\rightarrow$ (R), (D) $\rightarrow$ (S)

19. 90° **20.** 5 **21.** 8 **24.** $\sqrt{5}$ **25.** 4